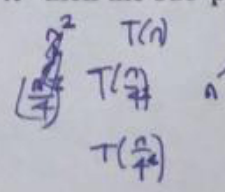
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Question 1: MCQs To be answered in (15) Minutes [ ]/ 5 Marks

1	2	3	4	5	6	7	8	9	10
B	B	D	D	A	D	B	D	B	A

1. Which of the following Big O notations represents the best (most efficient) time complexity?
   A. $O(n)$
   B. $O(1)$
   C. $O(n^2)$
   D. $O(log\ n)$
2. Which of the following algorithms has a time complexity of $O(n)$
   A. Heap Sort
   B. Radix Sort
   C. Merge sort
   D. Binary Search
3. The big O-notation of the expression $f(n) = 2^n + 6n^2 + 3^n$ is
   A. $O(2^n)$
   B. $O(n^2)$
   C. $O(n)$
   D. $O(3^n)$
4. What is the asymptotic relationship between $\log n^{\log 6}$ and $\log 6^{\log n}$.
   A. $\log n^{\log 6}$ is $O(\log 6^{\log n})$
   B. $\log n^{\log 6}$ is $\Theta(\log 6^{\log n})$
   C. $\log n^{\log 6}$ is $\Omega(\log 6^{\log n})$
   D. All of the above.
5. Which of the following is a stable sorting algorithm?
   A. Merge sort
   B. Quicksort
   C. Selection sort
   D. All of the above
6. Consider the recurrence: T(n)= T(n) + 1, using the Master method, choose the correct answer.
   A. Apply case 1 and $T(n) = \Theta(1)$
   B. Apply case 2 and $T(n) = \Theta(\log n)$
   C. Apply case 3 and $T(n) = \Theta(1)$
   D. Master method cannot be used.

   b = 1 ! ×
7. Suppose you have the following recurrence relation $3T(n/4) + n^2$ then the sub-problem size at level i:
   A. $\log_6 n$
   B. $n/4^i$
   C. $\log_4 n$
   D. None of the above



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8. Given the following two functions: $\sqrt{n}$ and $n^{\sin(n)}$, where $n > 0$, then:
   A. $\sqrt{n} = O(n^{\sin(n)})$
   B. $n^{\sin(n)} = O(\sqrt{n})$
   C. $n^{\sin(n)} = \Omega(\sqrt{n})$
   D. All answers are incorrect.

9. Consider a binary min-heap implemented using an array. Which one of the following arrays represent a binary min-heap:
   A. 25, 12, 13,16,8, 10, 14
   B. 8, 10, 14, 13, 12, 25, 16
   C. 8, 12, 16, 13, 10, 8, 14
   D. 8, 14,12, 13, 10, 8, 16

   B; 8, 10, 14, 13, 12, 25, 16

10. Counting sort performs ..... Numbers of comparisons between input elements.
   A. 0
   B. n
   C. nlogn
   D. n^2

Question 2: ***Fundamentals of Algorithms Analysis*** To be answered in (20) Minutes [1/ 7 Marks

1. Show that $\mathbf{f(n) = 3n^2 + 2n + 4}$ is $\Theta(n^2)$ and find the constants $c1$, $c2$ and n0 ? [2 marks]

O notation: $3n^2+2n+4 \leq 3n^2+2n^2+4n^2$

$\leq 9n^2$ $c_1 : 9$

Ω notation: $3n^2+2n+4 \geq 3n^2$ $c_2 : 3$

Θ notation:

$\therefore f(n): \quad 3n^2 \leq 3n^2+2n+4 \leq 9n^2 \quad \forall n \geq 1$

$f(n) = \Theta(n^2)$ why not $c_2 = 1$ (1.5)

$c_1 = 9$, $c_2 = 3$, $n_0 = 1$

2. Sort all the functions below in increasing order of asymptotic (big-O) growth. [2 marks]

$\mathbf{5^n}$, $\mathbf{n^{\log n}}$, $\mathbf{4 \log n}$, $\mathbf{n^4}$

$4\log n \leq n^4 \leq n^{\log n} \leq 5^n$

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3. Express the function $n^3/1000 + 100n^2 - 100n + 3$ in Ω-notation. [1 mark]

$f(n)$

$cn^3 \leq \frac{n^3}{1000} + 100n^2 - 100n + 3$ Polynomial

$c \leq \frac{n}{1000} + 100 - \frac{100}{n} + \frac{3}{n^2}$

This holds for $c=1$, $\forall n \geq 1$ $\quad n_0 = 1$ $\quad \therefore f(n) = \Omega(n^2)$

4. What is the time complexity of the following algorithm in the most precise Big O? [2 mark]

```
Algp(n)
  for i = 1 to n
    for j = 1 to 2^i
      Print(j)
```

$n \quad 2^n - 1 \quad n \times 2^n = O(2^n)$

$O(n2^n)$ ~~$= O(2^n)$~~ ✓

Question 3: *Recurrence* To be answered in (25) Minutes [ ] / 7 Marks

1. Using the recursion tree method to solve the following questions for the recurrence below:

$$T(n) = \begin{cases} 4T\left(\frac{n}{2}\right) + n, & n > 1 \\ 1, & n = 1 \end{cases}$$

A. Draw the recursion tree based on the recurrence. [2 mark]

Call Nodes # Sum

$T(n)$ $\quad 1 = 4^0$ $\quad n$

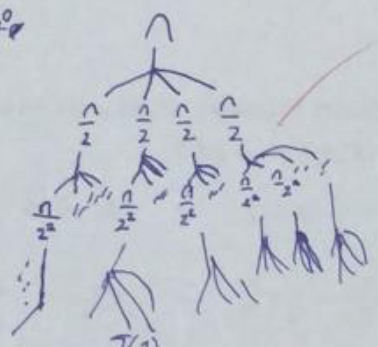
$T(\frac{n}{2})$ $\quad 4^1$ $\quad 4\frac{n}{2} = 2n$ ✓

$T(\frac{n}{2^2})$ $\quad 4^2$ $\quad \frac{16n}{4} = 4n = 2^2 n$ ✓

$\vdots$ Last Level

$T(\frac{n}{2^k}) = T(1)$ $\quad 4^k$

$k = \log_2 n$



$2^k n = 2^{\log_2 n} \times n$

$= n^{\log_2 2} \times n$

$= n^2$

B. Determine the number of levels in the tree and the cost at each level. [0.5 mark]

Number of levels $= \log_2 n + 1$ | Cost at level $i = 2^i n$

C. Determine the number of leaves in the last level. [0.5 mark]

$4^k = 4^{\log_2 n} = n^{\log_2 4} = n^2$

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D. Calculate the total cost (*Big-O*) of *T(n)* of all levels. Show all the details. [2 mark]

Total cost (Tc) = $L_c + I_c$ (L_c: Leaves Cost / I_c: Internal nodes cost)

$L_c = T(1) \times 4^{\log_2 n} = n^2$ → $n \times n^{\log_2 2} = n^2$

$I_c = n[(2)^0 + (2)^1 \dots + (2)^{\log_2 n - 1}] = n\left[\frac{2^{\log_2 n} - 1}{2 - 1}\right] : \frac{n2^{\log_2 n} - n}{1}$

$T_c = n^2 + n^2 - n \quad \therefore T(n) = O(n^2)$

2. Solve the recurrence below using Master method [2 marks]

$$T(n) = 8T(n/2) + n^3$$

$a = 8 \quad b = 2 \quad f(n) = n^3$

$n^{\log_2 8} = n^3 \rightarrow f(n) = \Theta(n^3)$

Use second case:

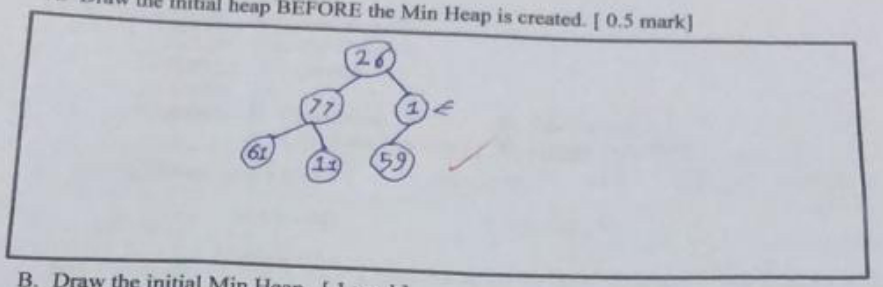
$$T(n) = \Theta(n^3 \log n)$$

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Question 4: *Sorting* To be answered in (25) Minutes [ ] / 6 Marks

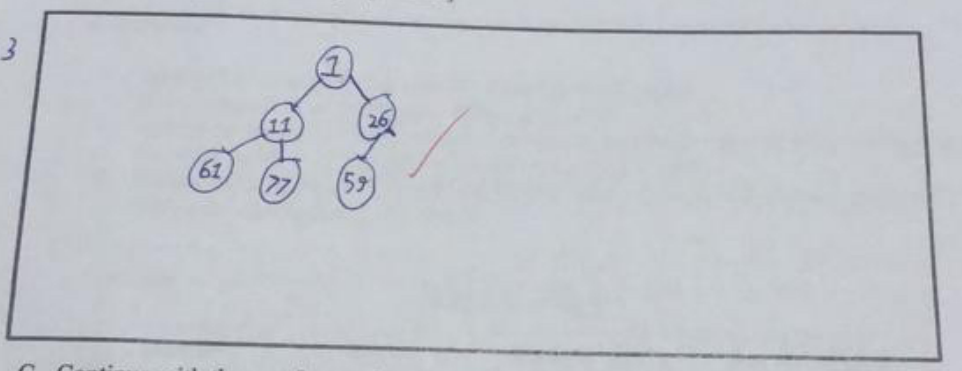
1. Given the following file sizes: 26, 77, 1, 61, 11, 59, sort them in descending order using Heapsort and answer the following questions:

   A. Draw the initial heap BEFORE the Min Heap is created. [0.5 mark]



   B. Draw the initial Min Heap. [1 mark]

$\lfloor \frac{6}{2} \rfloor = 3$



   C. Continue with the sort for one iteration PAST the initial Min Heap – i.e., you pop off 1 from the root. [1 mark]

   1st iteration:

11	59	26	77	61	1

   D. Now restructure the MinHeap after 11 is popped – what is the new root? [0.5 mark]

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DNA is a linear molecule made up of four types of subunits called nucleotides which are labeled A (Adenine), C (Cytosine), G (Guanine), and T (Thymine). Given the following four-letter DNA Sequences, ACGG, AAAA, TCCA, AGCT, CAAT, build a dictionary using Radix Sort, which can easily and quickly sort DNA sequences. [3 marks]

Original list	Sorting			
ACGG	AAAA	AAAA	AAAA	AAAA
AAAA	TCCA	CAAT	CAAT	ACGG
TCCA	~~ACGT~~ ACGG	TCCA	TCCA	AGCT
AGCT	~~CAAT~~ AGCT	AGCT	ACGG	CAAT
CAAT	CAAT	ACGG	AGCT	TCCA